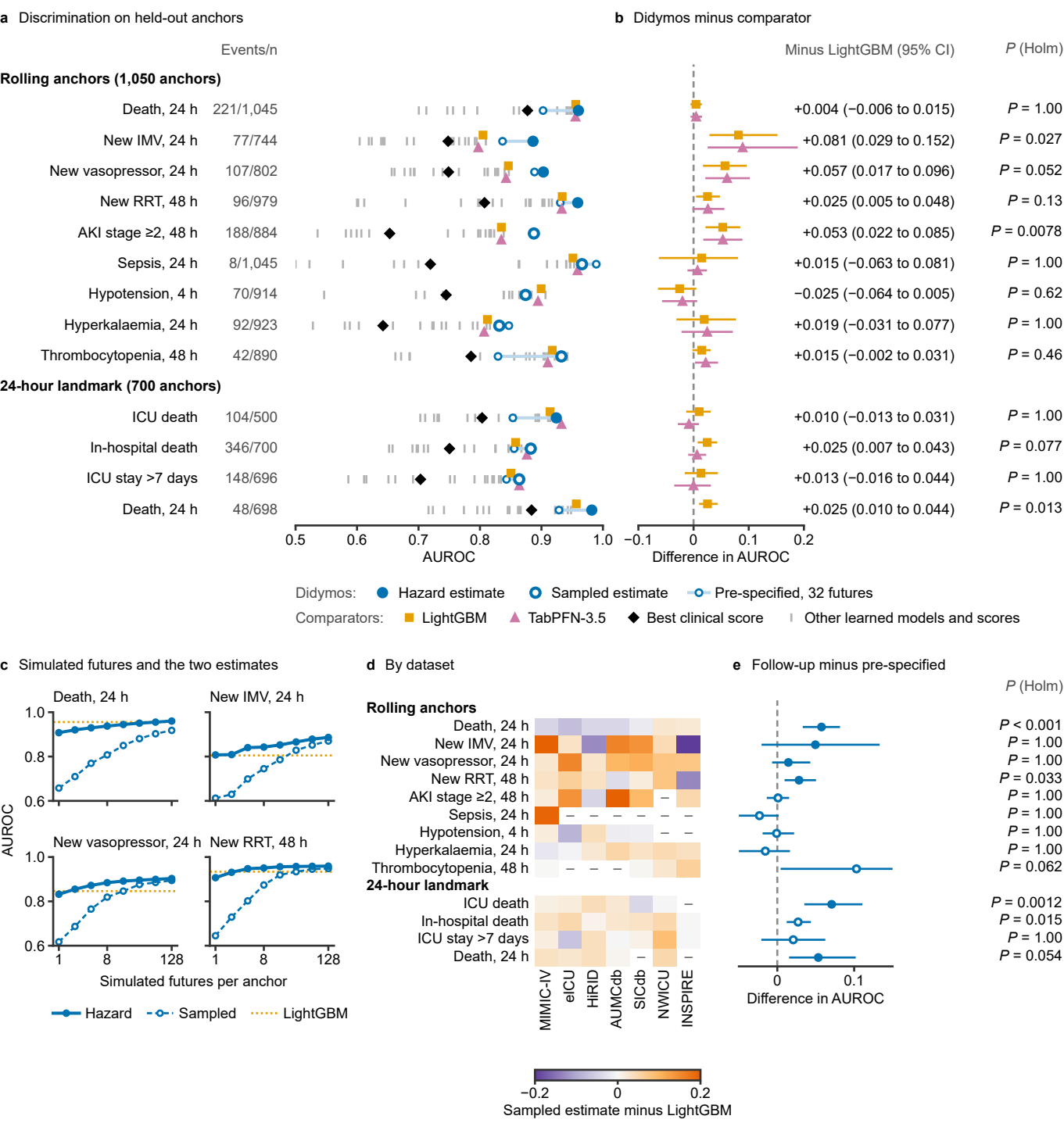

a Discrimination on held-out anchors
Events/n
b Didymos minus comparator
Minus LightGBM (95% CI)
P (Holm)
Rolling anchors (1,050 anchors)
Death, 24 h 221/1,045 +0.004 (−0.006 to 0.015) P = 1.00
New IMV, 24 h 77/744 +0.081 (0.029 to 0.152) P = 0.027
New vasopressor, 24 h 107/802 +0.057 (0.017 to 0.096) P = 0.052
New RRT, 48 h 96/979 +0.025 (0.005 to 0.048) P = 0.13
AKI stage ≥2, 48 h 188/884 +0.053 (0.022 to 0.085) P = 0.0078
Sepsis, 24 h 8/1,045 +0.015 (−0.063 to 0.081) P = 1.00
Hypotension, 4 h 70/914 −0.025 (−0.064 to 0.005) P = 0.62
Hyperkalaemia, 24 h 92/923 +0.019 (−0.031 to 0.077) P = 1.00
Thrombocytopenia, 48 h 42/890 +0.015 (−0.002 to 0.031) P = 0.46
24-hour landmark (700 anchors)
ICU death 104/500 +0.010 (−0.013 to 0.031) P = 1.00
In-hospital death 346/700 +0.025 (0.007 to 0.043) P = 0.077
ICU stay >7 days 148/696 +0.013 (−0.016 to 0.044) P = 1.00
Death, 24 h 48/698 +0.025 (0.010 to 0.044) P = 0.013
0.5 0.6 0.7 0.8 0.9 1.0
AUROC
−0.1 0 0.1 0.2
Difference in AUROC
Didymos: Hazard estimate Sampled estimate Pre-specified, 32 futures
Comparators: LightGBM TabPFN-3.5 Best clinical score Other learned models and scores
c Simulated futures and the two estimates
Death, 24 h
New IMV, 24 h
New vasopressor, 24 h
New RRT, 48 h
AUROC
1.0
0.8
0.6
1 8 128
Simulated futures per anchor
Hazard Sampled LightGBM
d By dataset
Rolling anchors
Death, 24 h
New IMV, 24 h
New vasopressor, 24 h
New RRT, 48 h
AKI stage ≥2, 48 h
Sepsis, 24 h
Hypotension, 4 h
Hyperkalaemia, 24 h
Thrombocytopenia, 48 h
24-hour landmark
ICU death
In-hospital death
ICU stay >7 days
Death, 24 h
MIMIC-IV eICU HiRID AUMCdb SICdb NWICU INSPIRE
−0.2 0 0.2
Sampled estimate minus LightGBM
e Follow-up minus pre-specified
P (Holm)
P < 0.001
P = 1.00
P = 1.00
P = 0.033
P = 1.00
P = 1.00
P = 1.00
P = 1.00
P = 0.062
P = 0.0012
P = 0.015
P = 1.00
P = 0.054
0 0.1
Difference in AUROC